



Daniela Breitman

Education

- 2026-2031 **Postdoc**, *University of Tokyo*, Physics.
2021-2025 **PhD**, *Scuola Normale Superiore di Pisa*, Physics, supervised by Andrei Mesinger.
2020-2021 **MSc**, *University of Toronto*, Department of Physics, supervised by Keith Vanderlinde.
2017-2020 **BSc**, *McGill University*, Montreal, HONOURS MATHEMATICS AND PHYSICS.

Doctoral Thesis

- Title Emulating the First Billion Years of the Universe
Supervisor Prof. Andrei Mesinger & Prof. Steven Murray

Master's Thesis

- Title A Study of the Signal Morphology from Fast Radio Bursts
Supervisor Prof. Keith Vanderlinde & Dr. Daniele Michilli

Bachelor's Thesis

- Title *Characterizing the Repetition Rate of CHIME Fast Radio Bursts*
Supervisor Prof. Victoria Kaspi

Organized Conferences

- September 2024 **Physics in the AI Era**, Pisa, Italy.
24-27 2024 Organising the conference included tasks such as inviting speakers, scheduling the talks, booking the conference dinner and catering, chairing the sessions, leading discussions, etc.

Seminars

- August 24-28 2026 **AstroAI Asian Network**, Taipei, Taiwan.
Invited to give a lecture and tutorials on generative modelling in astrophysics and cosmology.
June 20 2024 **University of Montreal**, Montreal, Canada.
[Sample Variance Denoising in Cylindrical 21-cm Power Spectra](#)
March 11 2024 **Tsinghua University**, Beijing, China.
[21cmEMU: An emulator of 21cmFAST summary observables](#)

Contributed Talks

- January 20-26 2024 **Workshop on AI-driven discovery in physics and astrophysics**, Tokyo, Japan.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- November 27-01 2023 **The Fourth National Workshop on the SKA Project**, Catania, Italy.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- September 11-15 2023 **6th Annual Global 21-cm Signal Workshop**, Trieste, Italy.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- March 6-10 2023 **Understanding the Epoch of Reionization**, Sesto, Italy.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- November 15-18 2022 **iid2022: Statistical Methods for Events Data – Illuminating the Dynamic Universe**, Lake Guntersville State Park, AL, USA.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- October 12-14 2022 **HERA Annual Meeting**, UC Berkeley, CA, USA.
[21cmEMU: An emulator of 21cmFAST summaries](#)
- October 17-20 2022 **5th Annual Global 21-cm Signal Workshop**, UC Berkeley, CA, USA.
[21cmEMU: An emulator of 21cmFAST summary observables](#)
- September 27-29 2022 **Cosmic Dawn/EoR SKA SWG Meeting**, Pisa, Italy.
[21cmEMU: An emulator of 21cmFAST summary observables](#)

Honours & Awards

- May-July 2020 **Characterizing the Repetition Rate of CHIME Fast Radio Bursts**, *Victoria Kaspi*, McGill University, Value: \$6,000.00.
- May-July 2020 **Solving the relativistic wave equation in FLRW cosmology using spectral theory on non-Euclidean spaces**, *Dmitry Jakobson, Linan Chen, Gantumur Tsogtgerel*, McGill University, Value: \$2,000.00.
- Summer 2019 **Summer Undergraduate Research Program (SURP)**, *Ue-Li Pen*, University of Toronto, Value: \$9,500.00 + \$2,000.00 FRQNT BPCA.
- Winter 2019 **Tomlinson Engagement Award for Mentoring (TEAM)**, McGill, Value: \$300.00.
- Summer 2018 **Undergraduate Summer Research Award (USRA)**, *Victoria Kaspi*, McGill University, Value: \$6,500.00 + \$2,000.00 supplement award FRQNT BPCA.

Technical Skills

- Proficiency with Linux (Ubuntu and Arch) operating systems.
- In-depth experience with pytorch, tensorflow, jax, matplotlib, numpy, scipy, pandas, bokeh, seaborn, emcee, multineest, ultranest, vi, Visual Studio Code, git, GitHub, astropy, pyyaml and h5py, and to varying extents django, plotly-dash and regex.
- Intimate knowledge of Python and to varying extents C, R, Java, HTML, CSS, Javascript and SQL.
- *Development and maintaining of Python packages:* [py21cmemu](#), [py21cmFAST-tools](#)

Languages and citizenship

- Native proficiency: English, French, Russian, Hebrew
- Good proficiency: Italian
- Citizenship: Canadian